7.1.1.1 Type of Leaf Spring

7.1.1.1 (a) Quarter Elliptic Leaf Spring
7.1.1.1 (b) Semi Elliptic Leaf Spring
7.1.1.1 (c) Three Quarter Elliptic Leaf Spring
7.1.1.1 (d) Full Elliptic Leaf Spring
7.1.1.1 (e) Transverse Leaf Spring

(a) Quarter Elliptic Leaf Spring

The figure 7.1.1.1 (a) shows the simple arrangement. The one end is fixed on the side member of the frame by means of U Clamp or I Bolt while the other end is freely connected to the front axle. In such springs the camber is provided on the upward side so that the leaves tend to straighten when the front axle beam is subjected to shock load. The cost and the weight of this type spring is low. This is also known as the cantilever type leaf spring and it was used in olden days.

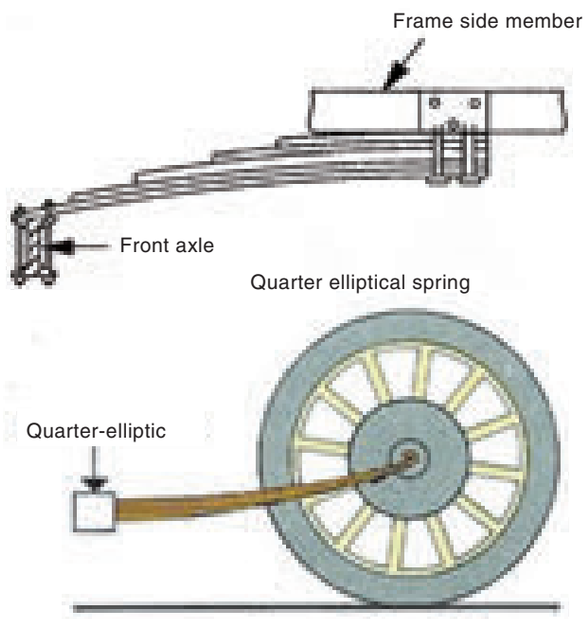


Figure 7.1.1.1 (a) Quarter elliptic leaf spring

(b) Semi-elliptical leaf spring

The simple construction is shown in the figure 7.1.1. (b). This type of arrangement is mostly used in automobile and it resembles the semi ellipse in shape. The leaf spring is made-up of a number of steel leaves. Each leaf is of a different length, but with equal width and thickness. The uppermost longest leaf having bushes at its two ends, is called the master leaf. The one end of the spring is rigidly fixed to the vehicle frame while the other end is attached with the help of shackle. The shackle accommodates the variation in the length while vehicle driving on rough terrains. This type of spring arrangement is requires less maintenance, easy to repair and it has long life.

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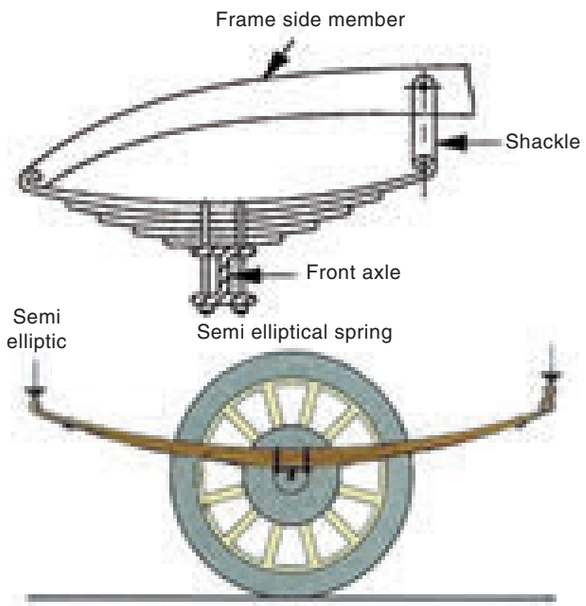


Figure 7.1.1.1 (b) Semi-elliptical leaf spring

(c) Three quarter elliptic leaf spring

The figure 7.1.1.1. (c) shows the arrangement. The three quarter elliptical spring is the combination of quarter elliptical spring and semi elliptical spring.

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